

Context Collapse in Long-Horizon Agents: Benchmarking Hierarchical Memory against RAG and Summarization

Ebaad Raheem^{a,*}, Danish Naseer^a, Ghulam Mustafa^a

^aDepartment of Information Technology, University of the Punjab, Gujranwala Campus, Pakistan

Abstract

Autonomous Large Language Model (LLM) agents that operate over long conversations face a critical weakness: *context collapse*, where factual details gradually drift and similar meanings interfere, weakening the agent’s ability to track its own state. This letter presents a carefully controlled benchmark of four memory designs; Baseline (full, untruncated history), Rolling Summary (repeated summarization of the dialogue), Retrieval-Augmented Generation (RAG), and a Hierarchical Memory system; evaluated across 800 fully reproducible conversational runs covering Legal, Medical, Tech, and Travel domains. Key facts are inserted at turn one and queried at five fixed recall distances using a two-level evaluation method that prevents the judge model from hallucinating. The Hierarchical architecture reaches an overall Factual Retention Rate (FRR) of 41.6%, performing competitively with the Baseline (40.6%) while using 34.8% fewer context tokens. RAG and Rolling Summary achieve 37.9% and 31.0%, respectively. Rolling Summary reduces context tokens by 60.5% but raises LLM API calls by roughly 205%, making it both expensive and harmful to performance. A strong *primacy effect* appears across all architectures as the conversation grows: at medium lengths (distances 14–19) retention becomes erratic (Baseline FRR falls to 39.0%), but then retention paradoxically jumps to 55.0% at maximum length, as attention locks tightly onto the start of the prompt. Domains differ sharply in fragility; the Medical domain collapses to a Baseline FRR of 12.0% and generates 192 explicit refusals under Rolling Summary, whereas Legal sustains 88.8%. These findings show that even with the full conversation history available, the Baseline essentially ties with hierarchical memory (40.6% vs. 41.6%, $p = 0.31$), yet domain collapse (Medical 12.0%) remains. Full context alone is not enough; memory structured at multiple levels offers greater resilience.

Keywords: context collapse, LLM agents, memory architectures, factual retention rate, retrieval-augmented generation, agentic state tracking, primacy effect

1. Introduction

LLM agents that engage in long, multi-turn conversations must keep precise factual state over dozens of turns; correctly recalling case numbers, drug dosages, or configuration settings introduced at the beginning and queried much later. Despite rapidly growing context windows, this state tracking remains brittle. The main failure is *context collapse*: the gradual loss of factual accuracy as an expanding dialogue is compressed with loss of detail. Every design choice; what to summarise, retrieve, or discard; adds semantic distortion that builds up over the conversation, leading not to a gentle decline but to a sharp drop, especially in domains dense with specialised terms.

This letter compares four cognitive architectures across 800 controlled runs and 4,000 evaluation data points from four professional domains. We map the efficiency frontier, uncover hidden compute costs of naive compression, and describe two understudied failure patterns: a universal conversational primacy effect and severe sensitivity to domain. Our results measure the

resilience that comes from hierarchically integrating working, episodic, and semantic memory layers, pointing to a promising direction for robust agent memory.

2. Related Work

Context Collapse and Long Contexts. Liu et al. (2024) formally described the “lost-in-the-middle” effect in LLMs: models recall information best at the boundaries of the context and badly in the middle. Our work extends this from static document retrieval to *dynamic state tracking in conversations*, where a fact’s effective position shifts continuously as new turns are appended.

Hierarchical and Retrieval-Based Memory. Robust agent memory needs organisation at multiple levels. MemGPT (Packer et al., 2023) implements the distinction between working, episodic, and semantic memory by managing a virtual context. However, because MemGPT relies on the LLM’s own reasoning to call “paging” tools, it is hard to benchmark the memory structure separately. Our method uses deterministic compression triggered by token count, which separates the memory design from reasoning artefacts, enabling a tightly controlled comparison with RAG and summarization. RAG (Lewis et al., 2020) takes a different approach by storing facts in an external vector index,

*Corresponding author.

Email addresses: bcs23033@pugc.edu.pk (Ebaad Raheem), bcs23006@pugc.edu.pk (Danish Naseer), gmustafa@pugc.edu.pk (Ghulam Mustafa)

but it struggles in conversational settings as topic boundaries blur and brief query wording drifts far from the densely stored evidence. Related work includes RET-LLM (Modarressi et al., 2023) for structured read-write entity memory, and long-term memory augmentation through separate side-networks (Wang et al., 2023).

Long-Horizon Conversational Agents. Recent studies on agent memory in extended dialogues have shown serious retention failures. Park et al. (2023) found that generative agents lose coherent state over hundreds of interactions without explicit episodic buffers. Maharana et al. (2024) benchmarked very long-term conversational memory and saw that even the best LLMs forget rapidly beyond 50 turns. Shinn et al. (2023) showed that verbal reinforcement can partly reduce memory loss, but at high computational cost. Our work stands apart by offering a controlled, multi-domain benchmark of four architectures under identical conditions, and by highlighting domain fragility and the efficiency trade-off as primary failure dimensions not covered by earlier evaluations.

3. Methodology and Experimental Setup

3.1. Controlled Synthetic Injection

We created 40 synthetic conversation scripts (10 per domain: Legal, Medical, Tech, Travel), each with 25 turns. Synthetic scripts were necessary to allow exact control over recall distance for measuring how memory fades with position, something impossible with unstructured dialogues. A block packed with several factual items is inserted at turn 1. Distinct facts from that initial block are later queried at five fixed probe points (turns 5, 10, 15, 20, and 25), corresponding to recall distances of 4, 9, 14, 19, and 24 turns. Queries use different paraphrases to prevent simple surface-form matching. With 5 independent repetitions for each strategy-script pair, the benchmark produces **800 conversational runs** and **4,000 individual evaluation data points**. The 25-turn length and dense entity interference deliberately create a tough stress test; therefore, the best retention rates stay around 40%, ensuring the benchmark measures memory resilience rather than task simplicity.

3.2. Memory Architectures

All four architectures use GPT-4.1-mini as the reasoning engine, so that the memory strategy is the only experimental variable.

- **Baseline.** The full, untruncated conversation history is supplied at every turn; this is the maximal raw-information strategy.
- **Rolling Summary.** After each turn, the entire history is replaced by a fixed-length summary generated by the LLM, introducing compounding loss of detail.
- **RAG.** Turns are split at sentence boundaries, embedded in a vector index, and retrieved by cosine similarity with a penalty that decays based on recency.

- **Hierarchical Memory.** Three layers are maintained: *Working Memory* (recent raw turns), *Episodic* records (structured entity-attribute pairs compressed from older working memory), and an immutable *Semantic* fact list (distilled from old episodes). Explicit semantic boundaries protect key facts from being lost during compression.

Hyperparameter Configuration. We set memory budgets so that all architectures worked with similar amounts of active context, separating the effect of the memory design from sheer token volume. RAG retrieved the top- k chunks ($k = 8$) using a combined score that weights semantic similarity at 70% and a recency bonus that falls off quickly at 30%: $\text{score} = 0.7 \cdot \text{sim}(q, \text{chunk}) + 0.3 \cdot \frac{1}{1 + \text{age}}$, where age is the number of turns since the chunk was added. Additionally, a separate buffer of the 6 most recent turns was always included regardless of the retrieval score. Rolling Summary was capped at 400 tokens per generation. The Hierarchical architecture triggered episodic compression when Working Memory reached 300 tokens, with episodic storage capped at 600 tokens.

3.3. Two-Level Evaluation Rubric

Level 1 performs deterministic exact substring matching of the target entity against the agent’s response. If that fails, **Level 2** calls a constrained LLM judge (GPT-4.1-mini) that is only allowed to output one of {CORRECT, PARTIAL, INCORRECT}. PARTIAL is given when the response contains the core entity but with minor inaccuracies (e.g., spelling errors, wrong unit, or missing qualifier) that still preserve the factual intent (Zheng et al., 2023). FRR counts both CORRECT and PARTIAL as successful retentions; explicit refusals are counted as failures.

4. Empirical Results and Discussion

4.1. The Efficiency Frontier

Table 1 and Figure 1 show a sharp, uneven trade-off between compression and accuracy. The Baseline consumes 1,242 context tokens per turn and averages 15.1 API calls per run (reflecting LLM calls triggered only on user-directed turns). The Hierarchical architecture cuts tokens by 34.8% to 810 while reaching the highest overall FRR ($41.6 \pm 1.5\%$). A paired t -test across repetitions shows that this overall margin over the Baseline ($40.6 \pm 1.7\%$) is not statistically significant ($p = 0.31$). However, looking only at overall averages hides different risk profiles: as detailed in Section 4.3, Hierarchical memory essentially ties with Baseline overall because it trades a little accuracy in highly structured domains (Legal) for preventing complete failure in dense domains (Medical). RAG further reduces tokens to 551 (a 55.6% reduction) at the same API cost as the Baseline (15.1/run), making it the most efficient choice on the token-accuracy frontier under tight token budgets.

Rolling Summary, however, represents an *inefficient trade-off*: its 60.5% token reduction (491 tokens) is entirely cancelled out by a roughly 205% increase in API calls (46.0/run) from extra summarization steps. It also delivers the worst FRR ($31.0 \pm 0.9\%$); a 9.6 percentage-point deficit compared to the Baseline; because repeated lossy summaries gradually reword, round off, or silently drop key factual values.

Table 1: Overall factual retention and efficiency per architecture. Token counts are mean context tokens/turn; API calls are total invocations per 25-turn run. FRR is reported with standard deviation ($\pm$) across 5 independent repetitions under lenient scoring (CORRECT+PARTIAL).

Architecture	FRR (%)	Tokens	API Calls	Token Δ
Baseline	40.6 $\pm$ 1.7	1,242	15.1	—
Hierarchical	41.6 $\pm$ 1.5	810	25.9	−34.8%
RAG	37.9 $\pm$ 1.4	551	15.1	−55.6%
Rolling Summary	31.0 $\pm$ 0.9	491	46.0	−60.5%

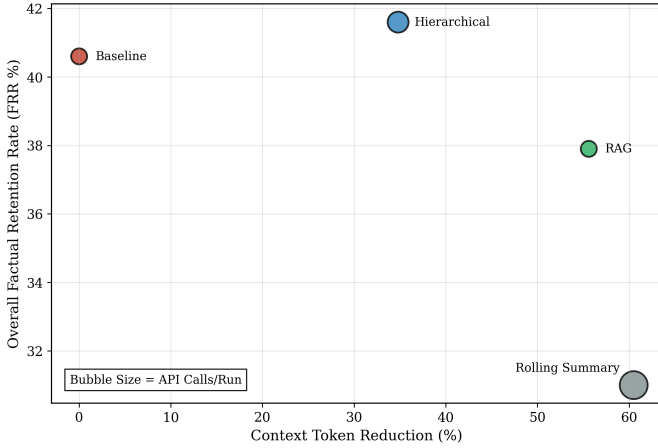


Figure 1: Efficiency frontier: context token reduction vs. overall FRR. Bubble area is proportional to API calls/run. Rolling Summary lies far outside the frontier because it loses accuracy and raises API calls by roughly 205%.

4.2. The Conversational Primacy Effect

Figure 2 shows how retention fluctuates across architectures as the conversation lengthens. At medium distances (14–19), mounting dialogue noise causes retention to bounce erratically, with the Baseline dropping to 39.0% at distance 19. Yet at the maximal distance of 24, retention paradoxically spikes to 55.0% (Baseline). This fits the known strong primacy bias of LLMs: when the context is very long, attention mechanisms latch firmly onto the absolute start of the prompt. While the Baseline spikes higher at the extreme boundary because it keeps the raw text, it suffers greater mid-dialogue instability. The Hierarchical architecture provides steadier mid-dialogue stabilisation, trading a little accuracy on rare edge cases for preventing severe failure in dense domains (Section 4.3), which yields a superior overall FRR.

4.3. Domain Brittleness and Context Collapse

Figure 3 reveals striking domain fragility, with a 76.8 percentage-point gap between Legal and Medical FRR. Legal entities (docket numbers, statutory citations) are syntactically distinct and stable, leading to a Baseline FRR of 88.8% and Hierarchical of 84.8%. Travel (Baseline: 31.2%) and Tech (Baseline: 30.4%) sit in the middle: their moderately distinctive entities can be confused by paraphrasing but still keep enough regularity for partial credit. The Medical domain shows the most severe collapse: Baseline FRR 12.0%, and the best-case Hierarchical

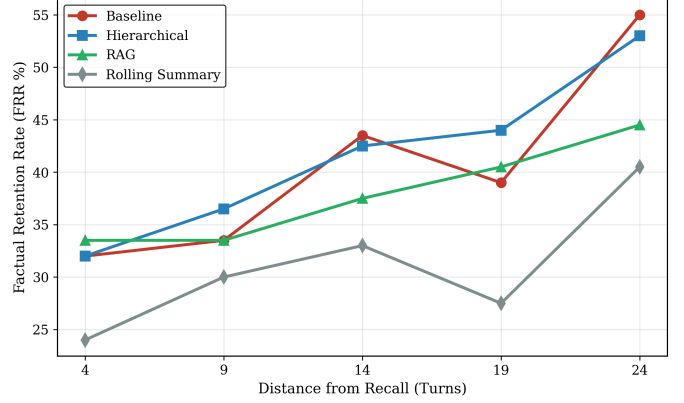


Figure 2: FRR across conversational recall distance under lenient scoring. Mid-dialogue distances are volatile (Baseline dropping to 39.0% at distance 19); at the maximal distance of 24, attention locks onto the prompt boundary and retention jumps to 55.0%. Hierarchical memory offers steadier stabilisation in the middle of the dialogue.

only 17.2%. Dense clinical terms; drug names, dosages, administration frequencies; appear alongside semantically similar but factually different neighbours, causing summaries to mix them up or overgeneralise. Under Rolling Summary, this leads to 192 explicit “I don’t know” refusals in Medical, versus only 53 in Legal; clear evidence that repeated lossy summaries destroy pharmacological precision to the point where the agent recognises, but cannot resolve, its own uncertainty. While Hierarchical memory’s structured Semantic fact list is the only tested mechanism that meaningfully eases Medical collapse (bringing FRR to 17.2%), the absolute failure rate remains far too high. This suggests that even when facts are preserved, dense clinical language still overwhelms the LLM’s ability to retrieve and reason, causing failure at generation time.

Limitations. Although this synthetic benchmark successfully isolates how memory decays with position, several limits affect how far the findings can be generalised. First, the small number of repetitions ($N = 5$) weakens the statistical power of comparative claims, so small performance differences must be interpreted cautiously. Second, we rely on a single model (GPT-4.1-mini) for both agent reasoning and Level-2 evaluation, making future validation across different models necessary to rule out architecture-specific or self-evaluation biases. Furthermore, while our baseline RAG uses standard, fair hyperparameters, thorough ablation studies and testing on natural, unstructured conversational noise are still needed to fully map the failure modes of these architectures.

5. Conclusion

This letter provides quantitative evidence that even under a tough 25-turn benchmark with dense entity interference, the Baseline (full history) does not outperform a token-efficient hierarchical memory (40.6% vs. 41.6%). Moreover, even with full context, the Medical collapse (12.0% FRR) confirms that simply expanding context windows cannot prevent context collapse. Architectural memory management is necessary. The

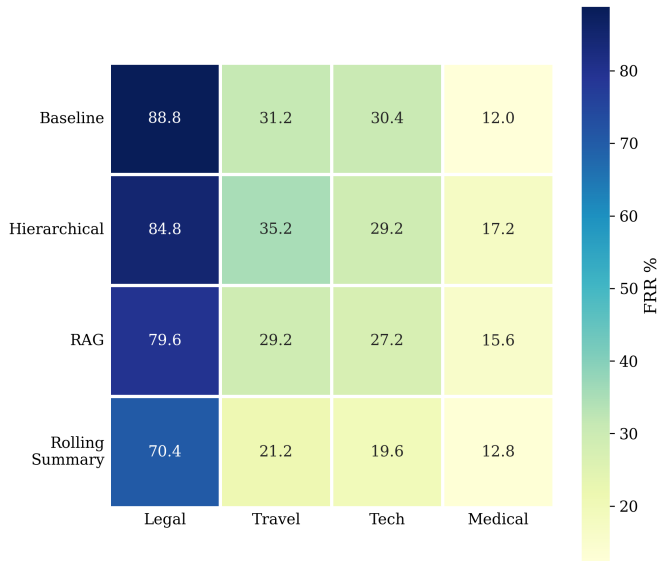


Figure 3: Per-domain FRR heatmap across all four memory architectures (values annotated per cell). The 76.8 percentage-point spread between Legal (peak: 88.8%) and Medical (floor: 12.0%) confirms that entity density is a first-order determinant of retention, independent of architecture. No strategy overcomes Medical collapse; only Hierarchical Memory meaningfully reduces it (17.2%).

Hierarchical Memory architecture; which achieves statistically equivalent FRR (41.6% vs. 40.6%, $p = 0.31$) through structured semantic preservation at multiple levels; offers factual retention comparable to the full-history Baseline while using 34.8% fewer tokens and avoiding the domain collapse that full context alone cannot fix. Rolling Summary is an inefficient trade-off: a roughly 205% rise in API calls and a 9.6 percentage-point FRR deficit compared to the Baseline rule it out for precision-critical deployments. The strong primacy effect, with late-dialogue recall spikes amidst mid-dialogue volatility, and the severe Medical fragility (12.0% Baseline FRR; 192 refusals under Rolling Summary) confirm that memory designs that adapt to domain and lock onto key entities are a vital area for future research.

Future work will explore compression policies that adjust to entity density and ensemble strategies that combine RAG and Hierarchical Memory to cover each other’s weak spots. Additionally, future studies should include statistical significance tests over larger repetition sets and evaluations across different models (e.g., Llama 3, Claude 3) to confirm that these architectural resilience patterns hold beyond the GPT-4.1 family.

Data and Code Availability. To ensure full reproducibility and transparency, all synthetic conversational datasets, the automated evaluation scripts, and the complete benchmark execution framework will be made publicly available in an open-source GitHub repository upon acceptance of this manuscript.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the authors used AI tools (Gemini, Claude) to refine the language, check grammar, and format the manuscript. After using these tools, the authors reviewed

and edited the content as needed and take full responsibility for the content of the publication.

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